Bleaching is another important pre-treatment next to scouring, performed on cotton fibres. This treatment is given to decolourize the natural colouring matter present in the cotton fabrics and impart a pure white colour. It increases the ability of the textile materials for dyeing and printing by removing any traces of colour present in it. Normally oxidative bleaching action is performed in the industries on cotton fibre substrates. Though a number of bleaching agents are available in the chemical market, few bleaching agents are used extensively. Calcium hypochlorite (CaOCl), Sodium hypochlorite (NaOCl) and Hydrogen peroxide (H_2O_2) are the most frequently used bleaching agents in the conventional cotton processing units.

Hydrogen peroxide is considered a universal bleaching agent, as it is suitable for all sorts of textiles. It is stable at neutral and value is near neutral pH. The pH of hydrogen peroxide solution can be modified based on the suitability. Processing pH of cellulosic is between 9 and 11.5; proteins 2.5 to 6.0 and for synthetics it is near neutral acidic pH.

In order to bleach cotton, the pH of hydrogen peroxide solution is maintained at around 11. When alkali is added the stability of hydrogen peroxide is reduced and decomposed at a fast speed. To control the rapid decomposition a stabilizer is added to the solution. The ingredients added during peroxide bleaching are normally; hydrogen peroxide (bleaching agent, 1-3% owm), sodium hydroxide or carbonate (bleaching promoters, 0.25 to 1.0% owm), sodium silicate (buffer or stabilizers, 0.5 to 1.0% owm)

$$H_2O_2 \longleftrightarrow H^+ + HO_2^-[\text{Stable}]$$

$$HO_2^- \longleftrightarrow [OH]^- + (O)[\text{Unstable}]$$

In the alkaline condition the instability of peroxide is continued by the concentration. The liberation of nascent oxygen is utilized for the oxidation reaction in a controlled manner by selecting the stabilizers to get uniform application.



Figure 1.1 Scouring Bleaching machine

1.3.3 Calendaring

It is a mechanical process that finishes the fabric, by passing it between sets of rollers and applying heat and pressure. The outside of the rollers can be smoother or engraved to give the perfect finish to the fabric, the structure of the rollers varies from hardened chromium plated steel to elastic thermoplastic rollers. By varying the rollers, adding any additional chemical treatment and temperature, a variety of calendared finishes can be produced like glazed or moiré fabrics.

Objectives

Calendaring is done for many purposes but the main objectives are:

- To give softening to the face side of the fabric
- To increase fabric lustre or glaze
- To give silk like appearance

XII TDD_English version CHAPTER 01.indd 4 1/13/2020 2:40:59 PM